

ChemWorld: Controlled Chemical Worlds for Studying Autonomous Scientific Discovery

Jiangjie Qiu^{1,*}, Yijun Li^{1,*}, Yaotian Yang^{1,*}
Honghao Chen¹, Wentao Li¹, Xiaonan Wang^{1,†}

¹Beijing Key Laboratory of Artificial Intelligence for Advanced Chemical Engineering Materials, State Key Laboratory of Chemical Engineering and Low-Carbon Technology, Department of Chemical Engineering, Tsinghua University, Beijing 100084, China

*Jiangjie Qiu, Yijun Li and Yaotian Yang contributed equally.

†Correspondence: wangxiaonan@tsinghua.edu.cn

Abstract

Autonomous scientific discovery requires converting limited experiments into knowledge that remains useful under new interventions. Evaluating this ability in real laboratories is difficult because the underlying mechanisms are incompletely known and agents choose the evidence on which they are later judged. We introduce ChemWorld, a programmable framework for constructing chemical worlds with evaluator-known process laws, persistent experimental state, explicit resources, and controllable public information. Composable process and observation models, qualified execution, and exact environment replay connect agents’ scientific claims to the experiments that precede them. Using this framework, we study 225 autonomous research campaigns across six chemical system families, with opaque, aligned, or misindexed prior information and system-specific discovery, characterization, or constrained delivery objectives. Agents freely select experiments and formulate mechanisms before predicting outcomes at withheld conditions. The results expose a separation between useful operation and predictive reliability. In electrochemistry, optimization assignments improve recommendation retests in 26 of 30 matched comparisons but improve score prediction in only 14. Increasing the experimental budget substantially reduces prediction error in electrochemistry and phase partitioning, while particle-quality prediction and uncertainty calibration remain difficult in downstream processing. Prior information has no uniform performance ordering across tasks or response variables. These findings support evaluating scientific agents through distinct measures of operational quality, predictive accuracy, and uncertainty, with mechanistic accounts grounded in recorded evidence. ChemWorld provides an experimental basis for investigating how autonomous agents acquire, revise, and use scientific knowledge.

1. Introduction

An autonomous scientist must decide which experiments to perform and what the resulting observations justify. Finding one productive condition is useful, but it does not establish that the surrounding system is understood. A local operating rule may fail under a change of material, process history, or measurement regime. Conversely, an experiment with poor immediate performance may be valuable because it distinguishes competing explanations. Evaluating autonomous research therefore requires following both experimental achievement and the knowledge an agent can use beyond its chosen experiments.

Language-model agents can now plan chemical procedures, use specialized tools, and interact with automated laboratories (Boiko et al. 2023; Bran et al. 2024; Szymanski et al. 2023). These developments make the relationship between action and understanding an empirical question. An agent’s final performance combines the information it was given, the evidence it chose to acquire, and its subsequent interpretation. In a physical laboratory, the relevant mechanism is usually only partly known; repeating the same research process under a different prior is expensive, and equivalent starting states are difficult to ensure. A final task score alone cannot locate where an unsuccessful scientific account arose.

We present ChemWorld as an experimental instrument for this problem. Its worlds combine chemical-process models, instruments, persistent material states, and resource rules. The researcher controls private process laws and public information separately, while an agent interacts through laboratory operations and measurements. This separation enables matched studies of different information conditions in the same world. Complete execution records and exact environment replay preserve what was done, what was measured, and what remained available at each step.

We use the platform to study a single agent configuration across six system families: electrochemical conversion, phase partitioning, reaction and thermal processing, aqueous equilibrium, crystallization, and purification. The tasks reflect their systems. Reaction and electrochemical studies separately assign discovery and optimization objectives; equilibrium and partitioning emphasize characterization; downstream processing requires quality-constrained delivery. Each campaign ends with a free-form mechanistic account, withheld-condition predictions, and an evidence-focused retrospective interview. Where an operating recommendation is required, it is sealed before this assessment and independently retested.

The principal empirical finding is that operational quality, prediction, and uncertainty can have different orderings. More experimental resources improve several prediction endpoints, but do not make every response variable reliable. Aligned information is useful in some settings without dominating across goals, worlds, or metrics. These results argue for evaluating scientific research as connected capabilities whose relationships must be measured, rather than inferred from one successful endpoint.

The paper makes three contributions: a composable, qualified instrument with separate physical and information controls; a common assessment sequence that preserves autonomous experiment selection and unrestricted mechanism expression across scientific commissions; and complete selected cohorts showing how operational and predictive outcomes diverge. Unfavorable comparisons and incomplete source campaigns remain part of the evidence.

2. ChemWorld as an experimental instrument

2.1 Constructing worlds with known laws and restricted observations

ChemWorld represents reaction, thermal, phase, separation, crystallization, distillation, continuous-flow, electrochemical, and observation processes as reusable components. Components declare their states, parameter domains, dependencies, and interfaces. Compatible declarations compile into executable worlds with legal operations, instruments, resource rules, and termination and evaluation conditions. Fifteen reference tasks illustrate this construction space. The broader research programme defines nine task families, including resource-limited characterization, continuous flow, and distillation; the present autonomous-agent evidence covers six of them. Task families can share physical components and are not independent chemical laws.

Write a world as $\mathcal{W} = (W_{\text{pub}}, \theta)$, where W_{pub} specifies the public experimental interface and θ contains private material properties, process parameters, and initialization. The agent receives usable operation and instrument contracts and observes measurements generated through those contracts, without reading θ or inspecting the simulator implementation. Knowledge of the internal model belongs to the evaluator, not to the participant.

This design supports two separate interventions. A **world intervention** changes a private law while preserving a matched public interface. An **information intervention** changes the supplied description while holding the world fixed. The platform includes six matched demonstrations of private-law forks. The three-arm campaigns studied here use information interventions within each world; the fork demonstrations establish a platform capability and do not increase the agent-study sample size.



Figure 1: Controlled worlds and autonomous assessment. The evaluator controls physical laws and supplied information separately. Agents retain freedom over experimental actions and mechanism expression. Prediction follows a sealed report but retains the full research context; it is not a report-only transfer test.

2.2 Experiments have persistent consequences

Laboratory operations alter a persistent physical state and consume resources. Sampling can remove material; separation changes where material resides; thermal and cooling histories can affect later observations. Instruments expose specific measurements rather than an unrestricted state vector. Two campaigns with the same number of batches can therefore acquire different evidence and leave different material available for subsequent actions.

The runtime validates an action and commits its candidate transition only when the applicable checks pass. A rejected action retains the committed physical state while incurring the declared attempt consequences. Public observations and private evaluator records are distinct projections of the same execution. Records retain actions, transaction outcomes, measurements, resource changes, and termination. Exact replay reconstructs the bound environment and action sequence, including rejected actions. It does not regenerate the language model’s words or imply deterministic agent behavior.

A scientific claim can thus be traced to a particular intervention and measurement, and an apparent discovery can be checked against what the agent actually had access to. Environmental validity, scientific correctness, and agent success remain separate properties.

2.3 Qualification establishes an internal domain of validity

The platform’s frozen qualification covers reference worlds, generated compositions, component invariants, interfaces, and invalid inputs (Table 1). All counts refer to finite declared domains. They establish internal consistency and execution properties, not agreement with arbitrary wet-laboratory chemistry.

Qualification unit	Passed / tested	Property examined
Reference task-world units	64 / 64	Workflow, resources, replay
Boundary and categorical recipes	1,786 / 1,786	Declared-domain execution
Generated compositions	52 / 52	Construction beyond reference instances
Module probes; interface paths	32 / 32; 7 / 7	Component and interface consistency
Invalid-action probes	192 / 192	Registered rejection or rollback
Invalid compositions	7 / 7	Incompatible constructions rejected

Table 1. Frozen platform qualification. These are software and model checks, not autonomous research campaigns or independent scientific discoveries.

The generated cases include 18 topologies absent from the reference registry, eight new task-world identities using a registered reaction-distillation topology, and 26 additional coverage cases. Eight deterministic use cases retain 89 submitted actions, 88 committed transitions, one planned rollback, and eight final assays. A separate agent demonstration completes a 15-action lifecycle in a non-reference composition with exact replay. These studies show construction and execution beyond reference examples; they do not rank agents.

Each subsequent research block has its own public-interface, reference-execution, and replay checks. These verify that declared resources are delivered, information treatments leave physics unchanged, queries execute legally, and relevant response differences are accessible. No qualification rule requires the tested agent to succeed or an arm to win. Platform repairs require the affected qualification to be repeated; a historical release is not a certificate for every later execution surface.

3. Autonomous research design

3.1 System-specific commissions and matched information conditions

The study comprises 225 scheduled research campaigns across six system families and seven study blocks (Table 2). Every block uses five world instances and three information arms. The same physical world, available operations, and withheld-condition questions are used across arms within a block. World identities and treatment labels remain evaluator-side.

Opaque supplies the public experimental contract and anonymous material identifiers, without the instance-specific dossier under test. It does not remove general chemical knowledge or information in tool descriptions. **Aligned** adds nominal information associated with the intended entities, parameters, or structural relations; it is not a complete simulator specification. **MisIndexed** applies the block’s fixed reassignment or mismatch to that information. It is a controlled alternative dossier, not arbitrary false text.

The treatment’s locus is distinct from its arm. Entity-level (E) interventions change associations between anonymous materials and descriptors. Parameter-level (P) interventions concern an effective parameter or local response claim; structural (S) interventions concern process relationships. Each E, P, or S study therefore contains Opaque, Aligned, and MisIndexed arms. The purification system abbreviation P is distinct from the parameter-prior label P.

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
EC / E: electrochemistry	Discovery; score optimization	12, 24	60	1,080 / 1,080
PA / E: phase partitioning	Explain phase allocation	12, 24	30	540 / 540
RX / P and S: reaction and thermal processing	Discovery; safety-constrained optimization	12	60	720 / 720
EQ / P: bounded equilibrium	Characterize effective responses	12	15	180 / 180
EQ / S: structural equilibrium	Explain response structure	12	15	180 / 180
C / E: crystallization	Deliver crystals under quality constraints	12, 24	30	539 / 540
P / E: purification	Recover product subject to purity	12	15	178 / 180
Total	Six system families		225	3,417 / 3,420

Table 2. Complete selected study scope. Each campaign has one agent realization. One C campaign completes eleven of twelve batches. One P campaign uses all twelve vessel starts but discards two vessels, leaving ten final assays. All campaigns have sealed mechanism, prediction, and reflection stages; 223 meet the full source-and-assessment specification.

EC agents vary materials, potential, current cap, duration, and legal operation sequences. Optimization targets the public balanced-efficiency score; discovery targets an explanatory and predictive account. RX agents control materials, catalyst loading, actual thermal history, measurements, and termination or quenching. Its parameter-prior study concerns a local temperature-response claim; its structural study concerns a reversible-pathway hypothesis. Both goals yield an operating recommendation, secondary to discovery where applicable.

PA asks how product is allocated between phases under changes in materials, phase volumes, and processing. EQ concerns amount, volume, concentration, staged additions, dissociation, and precipitation. Neither imposes an artificial yield-optimization objective. C seeks seed-excluded crystal recovery subject to purity at least 0.80 and fines fraction at most 0.50, with particles present. P seeks recovery from the original charge subject to purity at least 0.80. C recovery divides recovered product net of seed mass by the target product present before separation; it differs from P’s original-charge recovery and overall yield.

3.2 Free experimentation under finite resources

All campaigns use GPT-5.6 Sol with medium reasoning effort in independent sessions. Agents choose conditions, repetitions, intermediate measurements, and grouping. They may fit a model using a public numerical tool or describe mechanisms in other forms. No fixed equation family, candidate-mechanism menu, or mandatory per-batch reflection is imposed. An agent can plan several experiments before requesting results. A batch, tool operation, model response, and decision made after feedback are therefore different units.

A campaign permits 12 or 24 experimental batches, a matching allowance of additional instrument uses, and one final assay per completed batch. Additional measurements are optional and allocated freely within the public contract. Operations, material stocks, execution time, and computational allowances are separately

bounded. Larger-batch conditions also receive larger execution envelopes; they do not hold computation fixed. Agents cannot access the hidden simulator, repository, or external retrieval. Invalid actions and their resource consumption remain recorded.

The 24-batch condition is a fresh session, not a continuation after a 12-batch campaign’s assessment. Budget contrasts compare complete research realizations under different resource envelopes, not the isolated value of twelve extra observations. English outputs are requested throughout; retained RX and EQ assessment prompts include Chinese instructions requesting English answers. Prompt language and calculator budgets are documented in Appendix A rather than assumed identical across systems.

3.3 Reports, predictions, reflection, and recommendation retests

For EC, RX, C, and P, the agent selects a completed batch as its operating recommendation before the mechanism report. The evaluator executes the selected procedure separately in the same world with a different observation seed. A retest measures the procedure’s observed performance; it does not establish global optimality, repeatability across many trials, or robustness to new worlds.

Every campaign then follows this sequence:

1. **K1, mechanism report.** The agent gives a self-contained account in natural language, mathematics, or pseudocode, covering variables, relationships, experimental support, applicability, alternatives, and unresolved factors. No internal simulator vocabulary is required.
2. **Q, withheld-condition prediction.** After K1 is sealed, the agent predicts twelve fixed independent new batches, giving point estimates and intervals. Questions are selected before the source campaign and fixed across matched conditions. Laboratory access is disabled; no target values or scores are returned.
3. **K2, retrospective review.** After Q is sealed, the agent discusses tested claims, a proposed discriminating experiment, and unreliable or unused evidence. The shorter protocol has three questions; RX and EQ use seven themes.

Bounded EQ/P additionally seals a structured equilibrium supplement, yielding fifteen extra submissions. Canonical EQ/S uses unrestricted K1 reports, without a closed-set family-classification endpoint. A superseded EQ pilot with a different participant contract is excluded from the primary cohort.

The original context is retained through K1, Q, and K2. Predictions consequently test the source agent with its full research history, not what a new recipient can recover from its report alone. K2 cannot revise predictions, and retrospective statements do not establish what the agent believed during earlier experiments. Reports support interpretation; no uniform mechanism-correctness or primary LLM-judge score is claimed.

3.4 Outcomes, references, and analysis units

We report response-specific prediction error, interval coverage and width, and task outcomes separately. For campaign c and response m , point error is

$$\text{MAE}_{cm} = \frac{1}{12} \sum_{q=1}^{12} |\hat{y}_{cqm} - y_{qm}^{\text{ref}}|.$$

The reference target follows each block’s contract (Table 3). RX and EQ compare point predictions to five-repeat means and measure interval coverage across individual reference observations. EC and P use fixed seeded reference observations; PA and C use underlying pre-sampling response values. These different targets should not be pooled into a common accuracy or calibration score.

System	Prediction responses	Reference target	Interval level
EC	Six conversion/efficiency responses, including score	Seeded final observation	80%
PA	Organic and aqueous product fractions	Noiseless pre-sampling fractions	90%
RX	Yield, conversion, selectivity, byproduct signal, risk, score	Five-observation mean; coverage over observations	80%
EQ / P and S	Normalized pH, dissociation, precipitation signal	Five-observation mean; coverage over observations	80%
C	Net recovery, purity, size index, fines fraction	Noiseless pre-assay responses	80%
P	Purity, original-charge recovery	Seeded final observation	80%

Table 3. Every campaign predicts twelve conditions. PA fractions are complementary, and its two decision questions reuse the conditions. Reference repeats reduce observation noise but do not add agent sessions.

Campaign metrics receive equal weight within a reported condition. Goal contrasts match world, arm, budget, and locus; budget contrasts match world, arm, and goal. All scheduled arms and world-level variation are shown. RX and EQ macro errors average their own response metrics equally. The score-only RX goal contrast is an additional descriptive reanalysis aligning that endpoint with EC; the original macro comparison is also retained.

There is one agent realization per cell and five world clusters per system, with multiple conditions reusing each world. Batches, queries, metrics, and goals are correlated readouts. We report descriptive magnitudes and paired directions without treating queries as independent replicates or claiming population-wide significance. Infrastructure recoveries retain their original attempts and resource records; unfavorable scientific results are not replaced. Two source-budget shortfalls remain explicitly marked in the summaries. These agent studies are development evidence, distinct from the platform’s frozen qualification.

4. Results

4.1 Better operating recommendations do not imply better prediction

Electrochemistry provides the clearest separation between operational improvement and predictive accuracy. Across 30 matched goal comparisons, optimization yields a higher independent recommendation retest score in 26, but lower score-prediction MAE in only 14. Thirteen comparisons improve both endpoints; another thirteen improve the retest while worsening prediction (Figure 2). At 24 batches, mean retest score is 0.74792 under optimization and 0.54903 under discovery, while score-prediction MAEs are much closer, 0.10818 and 0.11222.

The pattern is not confined to one favorable error definition. Across EC’s other five prediction responses, optimization wins in 11 to 16 of 30 matched comparisons. Removing all comparisons involving infrastructure-recovered sessions leaves better retests in 15 of 17 pairs but better score prediction in only eight; seven pairs retain the better-retest/worse-prediction pattern. This smaller, less balanced sensitivity subset supports the descriptive pattern rather than estimating recovery’s causal effect.

RX establishes a boundary on the interpretation. Optimization improves retests in nine of 30 matched comparisons, score prediction in eight, and six-metric macro error in six. Five comparisons improve retests while worsening each prediction readout. Optimization is therefore neither a uniformly better operating

strategy across these tasks nor necessarily a tradeoff against prediction. Operational and predictive outputs must be measured separately: their relationship changes with the system and assignment.

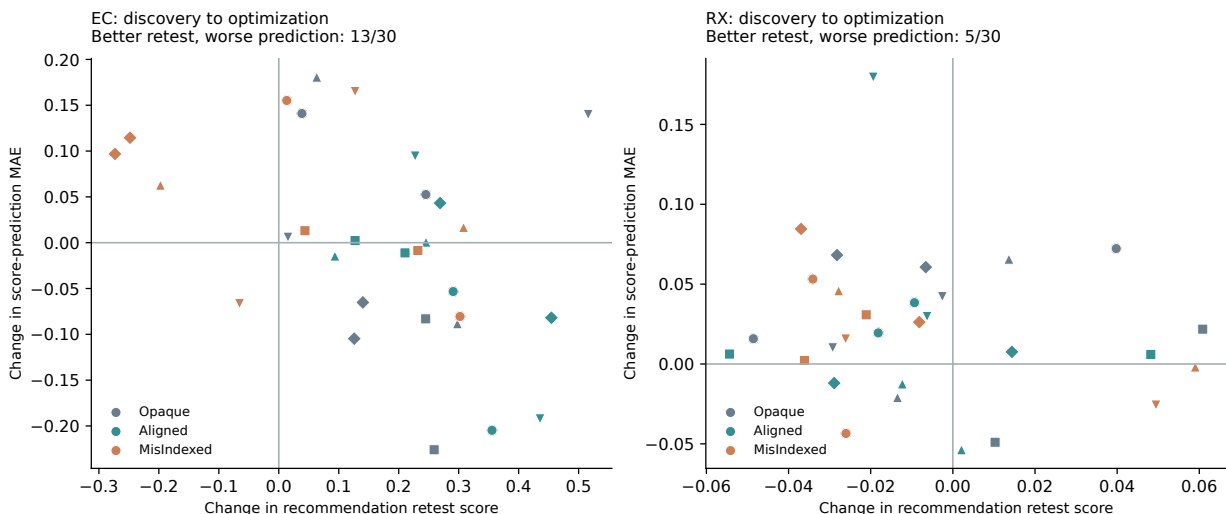


Figure 2: Goal contrasts with the same prediction endpoint. Points compare optimization with discovery in matched conditions; colors denote arms and shapes denote five worlds. Rightward means a better retest, upward means worse score prediction. Each panel contains 30 pairs clustered in five worlds. EC includes two budgets and RX two prior loci; scores and reference targets differ, so magnitudes are not pooled.

4.2 Additional resources improve prediction unevenly

Increasing the budget from 12 to 24 lowers mean score-prediction MAE by 35.4% for EC discovery and 39.3% for EC optimization. PA organic-fraction error falls by 71.3% (Table 4). Improvements occur in 11 of 15 matched EC discovery conditions, 12 of 15 EC optimization conditions, and 13 of 15 PA conditions. These include all planned conditions, including those where more resources yield worse prediction.

Assignment / response	MAE, 12 to 24	Coverage, 12 to 24	Width, 12 to 24
EC discovery / score	0.17378 to 0.11222	61.1% to 71.7%	0.19831 to 0.22007
EC optimization / score	0.17811 to 0.10818	55.6% to 73.9%	0.19935 to 0.20814
PA / organic fraction	0.08291 to 0.02379	78.3% to 98.9%	0.25122 to 0.18553
C / net recovery	0.10433 to 0.07937	67.8% to 71.7%	0.21313 to 0.21515
C / fines fraction	0.30999 to 0.27000	35.6% to 43.9%	0.32939 to 0.34789

Table 4. Budget comparisons average fifteen campaigns per budget. Nominal coverage is 80% except for PA’s 90%. C includes its eleven-of-twelve source. Campaign-level data retain all widths; Appendix B reports every metric and arm.

PA improves both point prediction and decision performance: its two decisions per campaign rise from 25 of 30 correct to 30 of 30, while intervals narrow. Coverage of 98.9% exceeds nominal 90%, so improved coverage does not imply exact calibration. EC’s 24-batch score coverage remains below nominal at 71.7% and 73.9%. Point accuracy and uncertainty reliability improve at different rates.

Crystallization shows why more resources do not remove every difficulty. Recovery MAE decreases from 0.10433 to 0.07937, but fines-fraction MAE remains 0.27000 after 24 batches, with only 43.9% coverage for nominal 80% intervals. Recommendations meet particle and purity criteria in 12 of 15 retests at each budget; ten at each budget also reach the initial recovery target of 0.10. Mean retest recovery changes from

0.42054 to 0.39939. More resources improve several predictions without a corresponding improvement in this constrained-delivery endpoint.

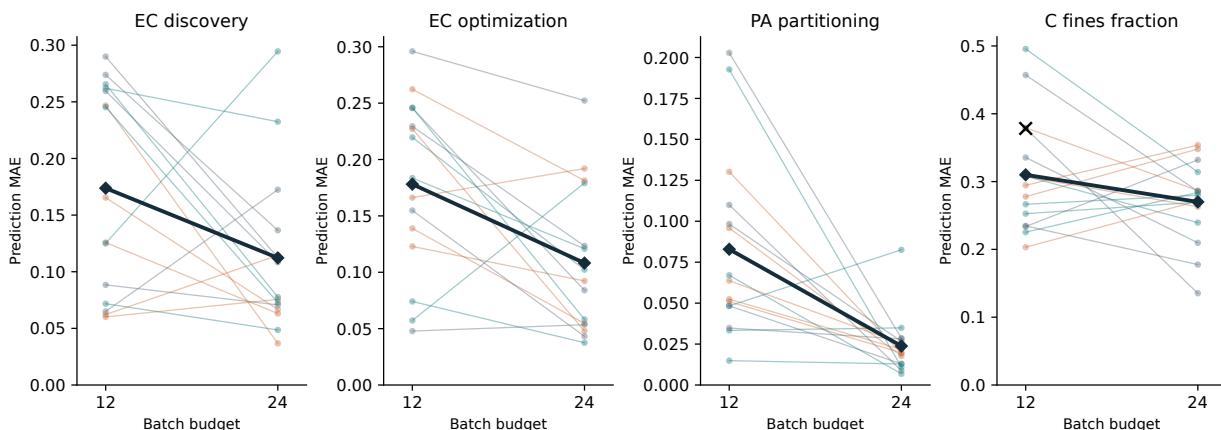


Figure 3: Budget contrasts. Thin lines connect the same world and arm across independent 12- and 24-batch sessions; dark diamonds show means. Colors follow Figure 2. The cross marks C’s source shortfall. EC panels predict score, PA predicts organic fraction, and C predicts fines fraction; vertical scales differ.

These comparisons enlarge the research envelope, including computation and allocation. They do not establish that twelve experiments are inherently insufficient or identify whether an agent needed broader coverage, replication, a better model, or better use of evidence already obtained. Retained histories permit those explanations to be examined without treating them as established by the budget contrast itself.

4.3 Prior information has no uniform performance ordering

The complete arm comparisons have no common ordering (Figure 4). In EC discovery, MisIndexed has the lowest mean score-prediction MAE at both budgets: 0.13206 and 0.07133, compared with Opaque’s 0.19534 and 0.12011 and Aligned’s 0.19393 and 0.14521. It beats Opaque in four of five worlds at each budget. Under EC optimization, Aligned instead has the lowest mean error and highest mean recommendation score at both budgets; its prediction beats MisIndexed in four of five worlds at each budget.

PA similarly resists a single ranking. Aligned has the lowest mean error at 12 batches, whereas MisIndexed has the lowest at 24. At 24, however, Aligned beats each comparator in three of five worlds; larger errors in the remaining worlds change the mean ordering. Paired directions and error magnitudes are both needed.

RX discovery changes ordering across loci. Parameter-prior macro errors for Opaque, Aligned, and MisIndexed are 0.10186, 0.05701, and 0.05457; structural-prior errors are 0.08404, 0.09209, and 0.10159. Bounded EQ/P favors Opaque on macro error (0.01214 versus 0.04314 and 0.03928), whereas canonical EQ/S means are closer (0.01408, 0.01590, and 0.01345), with different world-level orderings. Information content and query designs differ across these studies; comparing them does not isolate a causal advantage of one prior locus over another.

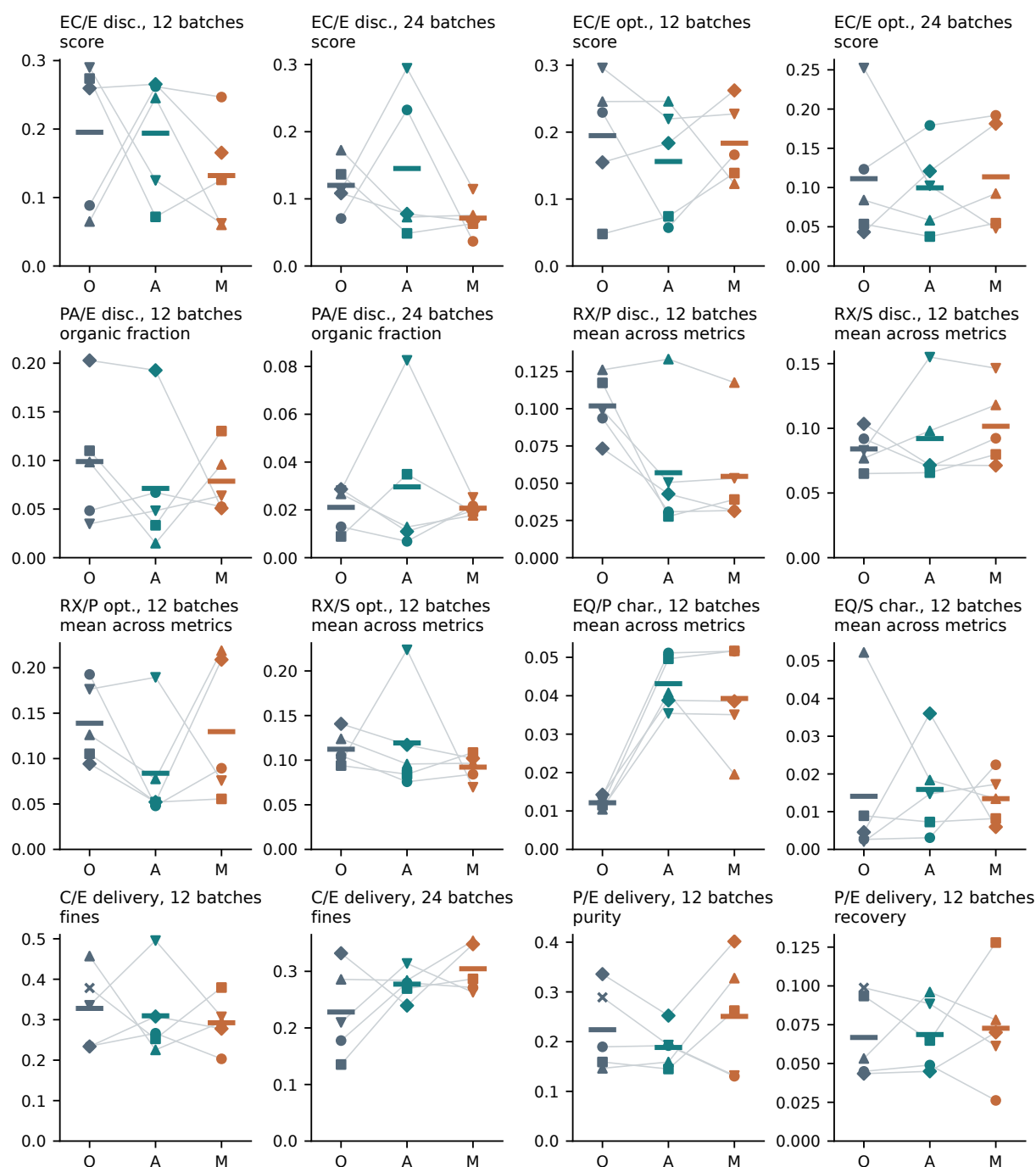


Figure 4: Prior information has no common ordering. Every panel includes five worlds and three arms: Opaque (O), Aligned (A), and MisIndexed (M). Gray lines connect worlds; colored bars show means. Crosses mark source shortfalls. RX and EQ use within-system macro errors; other panels name their response. Scales differ. Appendix B complements these readouts with all metric-specific arm means.

These outcomes do not establish that inaccurate information is intrinsically useful. A changed dossier can alter both experimental choices and subsequent inference, so arms acquire different data. With one stochastic realization per condition, the observed rankings cannot isolate reasoning conditional on identical evidence or prove that a particular supplied false claim was accepted, rejected, or corrected.

4.4 Quality constraints expose failures hidden by aggregate recovery

Purification makes the distinction between outcome dimensions concrete. Aligned recommendations achieve purity at least 0.80 in three of five worlds, compared with none for Opaque or MisIndexed. Yet Aligned’s mean original-charge recovery is 0.04455, compared with 0.12373 and 0.32105, respectively (Figure 5). The largest recovery is not the best delivery under the stated constraint. One Opaque retest has purity 0.7992, close to the threshold; continuous endpoints are essential alongside categorical counts, especially with one noisy retest per recommendation.

Predictive ordering also depends on the response. Aligned has the lowest mean purity MAE, while Opaque has the lowest recovery MAE (Table 5). Removing pairs involving the nonconforming P source retains the directions of these mean contrasts, with four world pairs remaining for comparisons against Opaque.

Arm	Purity MAE	Recovery MAE	Purity coverage	Recovery coverage	Eligible retests
Opaque	0.22396	0.06677	51.7%	56.7%	0 / 5
Aligned	0.18830	0.06865	53.3%	51.7%	3 / 5
MisIndexed	0.25078	0.07271	50.0%	53.3%	0 / 5

Table 5. Purification prediction and retest outcomes. Nominal coverage is 80%, against fixed seeded observations. All fifteen recommendations execute legally; execution validity is distinct from meeting the scientific objective.

Across arms and responses, nominal 80% intervals cover only 50.0% to 56.7% of targets. The paired-query analysis also finds thirteen of fifteen campaigns incorrectly predicting a purity change exceeding the declared small-effect threshold for the wash-staging pair. This is a shared error on one registered contrast, not evidence that all agents adopted the same mechanism or that wash staging can never matter. Together with C’s persistent fines errors, it motivates evaluating individual responses and interventions in addition to a recommended endpoint.

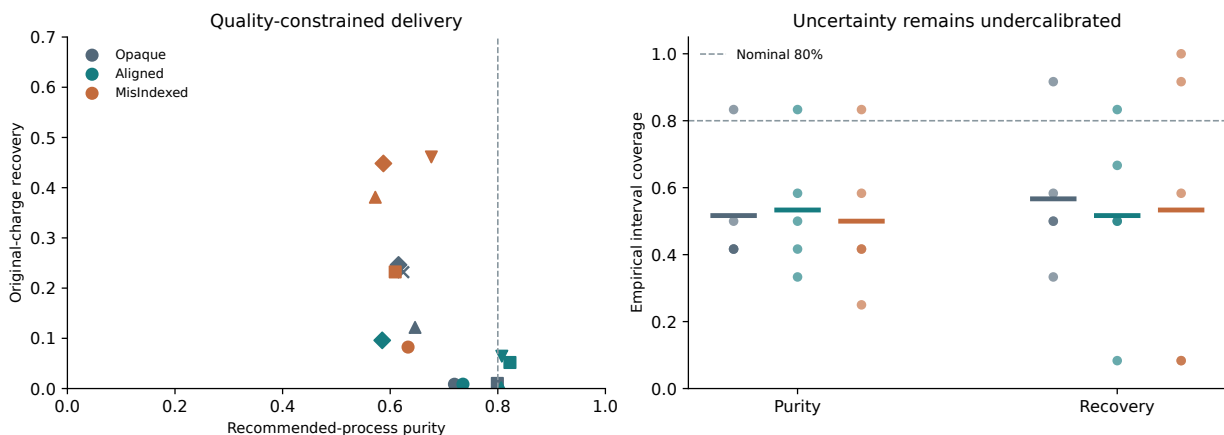


Figure 5: Purification delivery and uncertainty. Left: all fifteen recommendation retests, with the purity threshold dashed. Right: campaign-level interval coverage and arm means; the dashed line is nominal 80%. World markers and arm colors follow Figure 2; a cross marks the source shortfall.

4.5 A trajectory links limited exploration to an overbroad account

In one Opaque EC discovery campaign, the first experiment at +1.5 V performs poorly. The agent tests -1.5, -0.5, and -2.5 V next, then concentrates remaining experiments at -0.5 V while varying other conditions. Only one of twelve batches tests positive potential. Its sealed report describes cathodic conversion and treats

the positive direction as effectively inactive. Selectivity MAE is 0.01765 for negative-potential queries but 0.82634 for positive-potential queries.

This retrospectively selected example links observed coverage, an explanation extending beyond that coverage, and asymmetric prediction error; it does not estimate failure prevalence. Nor is counting positive-potential batches sufficient: two other 24-batch discovery campaigns each test two positive settings, yet their score MAEs are 0.23245 and 0.03676. Their tested locations and acquired evidence differ. An interpretation must follow actual interventions and claims rather than infer understanding from a coarse coverage count.

The record does not establish an immutable internal belief or causal information bottleneck. A systematic mechanism analysis requires consistent coding across all campaigns, including successful revision and justified unresolved alternatives. What this trace establishes is directly observable: which experiments occurred, what the report claimed, and where the predictions failed.

5. Discussion

5.1 Scientific evaluation needs more than a successful endpoint

The dissociations clarify what operating benchmarks measure. A recommendation tests whether an agent can propose a useful procedure. Withheld-condition prediction tests its ability to anticipate responses beyond its chosen experiments. Intervals test whether uncertainty accompanies those predictions. These outputs inform one another without being interchangeable. Their separation is consistent with the distinction between prediction loss and downstream decision quality (Elmachtoub and Grigas 2022), and here arises within autonomous experimental research.

This changes how failures should be described. Missing a purity requirement differs from executing an illegal operation. A useful crystal recipe can coexist with poor particle predictions. A nominal relation can be useful locally without being a complete law. Separate physical, informational, and evaluation interfaces preserve these distinctions rather than collapse them into one success score.

5.2 Evidence acquisition and interpretation need separate explanations

An agent may never perform the intervention needed to challenge a claim. It may instead acquire contradictory evidence and omit it, or recognize that observations leave several mechanisms possible. Behavioral records can distinguish these observable situations when claims and observations are explicitly aligned. A final error or retrospective self-description cannot do so alone.

Known simulator mechanisms help the evaluator design discriminating interventions and assess consequences. They do not make exact implementation recovery the only acceptable scientific explanation: multiple accounts can be observationally equivalent in a restricted domain. Calibrated non-identifiability may be preferable to an unsupported mechanism. Free-form reports should therefore be evaluated for supported relationships, scope, contradictions, and predictive consequences, while allowing unresolved alternatives.

This same-context protocol does not establish information loss caused by writing a mechanism report. Identifying that effect would require varying what a fresh recipient receives, such as the full record versus a report, with source evidence and prediction questions fixed. Separating a prior’s influence on experiment selection from its influence on inference similarly requires matched-evidence interventions. The platform enables these studies, but they are not current results.

5.3 Scope and limitations

The physical models are controlled abstractions qualified within finite domains, not validated representations of all chemistry. The six studied families share components; the remaining three task families lack comparable autonomous cohorts. One model configuration, five worlds per system, and one realization

per cell limit generalization across models and stochastic runs. Fixed reference queries address specified scientific questions, not every legal intervention.

Protocols also differ in prediction target, nominal interval level, prompt language, and computation envelope. Within-block comparisons preserve the relevant contract; absolute cross-system rankings would conflate these differences. Infrastructure recoveries and source shortfalls remain visible. Selected recovery and conformance sensitivities support particular comparisons without removing all sources of variation.

Numerical predictions are not a complete evaluation of mechanistic knowledge. Reports and reflection stages are retained, but systematic claim-level annotation is not complete. We distinguish quantitative findings from illustrative trajectories and hypotheses about their causes. The evidence supports separate evaluation dimensions; it does not yet establish a universal taxonomy or causal theory of information loss.

6. Related work

Autonomous chemistry systems demonstrate tool use, planning, and physical execution by language-model agents and robotic laboratories (Boiko et al. 2023; Bran et al. 2024; Szymanski et al. 2023). ChemWorld complements these efforts with controlled environments in which the evaluator varies process laws or supplied information and inspects complete experimental histories, making questions about research behavior accessible alongside costly physical investigations.

Reaction-optimization and experiment-planning benchmarks provide standardized objectives and conditions (Felton et al. 2021; Häse et al. 2021). Interactive digital chemistry environments expose sequential actions and chemical processes (Beeler et al. 2024). ChemWorld combines composable process models with separate public and private interfaces, persistent experimental consequences, and traceable assessment. This is a complementary emphasis, not a claim that existing environments lack every such capability.

Scientific-discovery environments increasingly evaluate hypothesis formation, experimentation, and model discovery, including DiscoveryWorld and BoxingGym (Jansen et al. 2024; Gandhi et al. 2025). Our contribution combines controlled chemical commissions with system-specific objectives and comparisons among information conditions, prediction, calibration, and recommendation retests. The empirical contribution lies in the observed relationships and the preserved evidence needed to inspect them.

7. Conclusion

ChemWorld makes the experimental foundations of autonomous scientific claims observable and controllable. Across the completed six-system study, better operating procedures, more accurate predictions, and reliable uncertainty do not consistently emerge together. Additional resources help several prediction tasks, and prior information changes outcomes, but neither has a uniform effect across commissions and responses. Evaluating scientific agents therefore requires recording what they investigated, what they claim, and what their accounts can predict, alongside the quality of the procedures they recommend.

Data and code availability

The platform release is available at [ChemWorld-Public](#). The autonomous-study analysis uses retained campaign summaries, sealed assessments, reference observations, and replay records. Accompanying review materials contain figure-generation code, campaign-level metric tables, and an evidence index connecting comparisons to source exports. The frozen platform release and later agent-study execution surfaces are distinct. This review does not imply that all later execution artifacts are already in the public release. Credentials and raw provider streams are excluded from distributable materials.

Appendix A. Protocol detail and evidence handling

A.1 System-specific predictions

EC's twelve conditions balance positive and negative potential while varying duration, materials, and current cap at a stated loading. Its six responses are selective product yield, electrochemical selectivity, Faradaic efficiency, transport efficiency, energy efficiency, and public score. Current is a magnitude cap, not guaranteed delivered current. PA predicts phase fractions before analytical sampling; sampling losses and complementary fractions must not be counted as separate discoveries.

RX's twelve paths examine time, temperature, catalyst, heating sequence, and quenching. Actual thermal telemetry matters: requested boundary temperature need not be realized reaction temperature. Five reference observations per condition are withheld from agents, with all source assessments sealed before reference-truth release. RX's 600 reference executions are separate from 720 source batches and 60 retests.

Each EQ block uses twelve questions and five observations per question, giving 300 reference executions. Bounded EQ/P examines effective aqueous relationships; canonical EQ/S supports response-shape analysis without requiring a named hidden family. An instrument's equilibrium diagnostic is not the agent's subjective confidence. The bounded study's effective-parameter supplement is additional to K1/Q/K2 and does not confer discovery credit for a parameter already supplied as a treatment.

C's twelve conditions form six pairs: material, seed dose, cooling history, prior thermal history, continued growth, and upstream loading. Its size response is $\min(d_{50}/250\text{ }\mu\text{m}, 1)$, a bounded number-weighted index rather than a diameter in micrometres. Fines are particles below 20 micrometres. Quality requires purity and fines criteria jointly, with particles present; recovery of 0.10 is an initial feasibility target, not an optimization ceiling. P's six paired interventions concern multistage recovery, including wash staging. Its original-charge denominator differs from C's crystallization-stage recovery.

RX and EQ coverage uses individual reference observations while point targets use means. EC and P use single seeded observations. PA and C prediction coverage concerns noiseless pre-sampling responses; C recommendation retests use observed assays. Apparent retest threshold crossings can therefore reflect noise. Macro errors weight a system's responses equally; normalized scales do not make responses physically or scientifically interchangeable.

A.2 Language, numerical tools, and stage budgets

Source briefs and requested reports are English. EC/PA and C/P assessment instructions are English; RX and EQ K1/Q/K2 instructions include Chinese text requesting English reports. Within-block arms have the same language contract. Seven-theme K2 reviews cover treatment claims, influential experiments, competing explanations, one proposed discriminating experiment, goal-related tradeoffs, unused evidence and uncertainty, and limitations of the recommendation or scientific account. Three-question versions combine these themes. The reviews occur after source experiments and sealed predictions; their length does not provide additional source experiments.

The earlier EC/PA block retains its saved eight-call diagnostic-calculator hard stop; later blocks disclose 128-attempt assessment allowances. An old EC parameter/structural supplement encountered the smaller limit and remains an execution failure outside the primary E cohort. The repair is not retroactively attributed to older sessions. Public calculation uses agent-accessible information, not hidden targets. Differences in numerical budgets and deadlines are execution contracts, not scientific findings.

A.3 Completion, recovery, and exclusions

The primary denominator is 225 scheduled campaigns, not provider attempts. All 675 K1/Q/K2 stages and 165 planned recommendation retests in EC, RX, C, and P are complete, plus fifteen EQ supplements. Final-assayed source batches total 3,417 of 3,420 planned. C's second-world 12-batch Opaque source has

eleven final assays. P’s fifth-world Opaque source starts twelve vessels, discards two, and completes ten final assays. Both remain in outcome tables with conformance indicators.

EC/PA has 68 first-attempt and 22 infrastructure-recovered campaigns. Recovery after transport or host failures retains earlier attempts and consumed resources. Missing assessment stages resume from saved context when the source is intact; fresh source replacements are recorded as such. Repeated question delivery and failed-attempt computation are not silently counted as an original clean attempt. EC goal sensitivity is reported in the main text. First-attempt-only EC-discovery, EC-optimization, and PA budget comparisons improve prediction in four of five, eight of eleven, and eight of ten pairs, respectively; these reduced subsets have uneven coverage.

The current P matrix follows a correction of prepartition inventory handling. Affected qualification and the entire agent block were restarted under the corrected contract; earlier apparently successful cells were not selected into the new matrix. All current source and recommendation trajectories pass inventory, resource, and replay checks. Its two lawful discards are a source-assay shortfall, not an inventory failure. The fixed purity-direction rule treats changes of 0.02 or less as small; a numerical tolerance of 10^{-12} corrects floating-point equality handling. Six direction classifications changed after this correction, with originals preserved. Predictions, point errors, intervals, retests, and denominators were unchanged.

Historical single-world EC parameter/structural supplements, the closed-set EQ/S pilot, early C/P development runs, qualification recipes, references, replays, and interrupted attempts do not enlarge the denominator. Their records remain separate. Frozen platform qualification is also a separate evidence programme. These counts describe execution and analysis scope; they are not 225 independent chemistry replications.

A.4 Mechanistic interpretation

The illustrative trace is the second-world, 12-batch Opaque EC discovery campaign; its sparse-positive-coverage counterexamples are first-world, 24-batch Aligned and MisIndexed discovery campaigns. They are retrospectively selected examples, not a coded prevalence estimate.

A systematic analysis should attach each mechanistic claim to its experimental support, contradictory observations, stated domain, and corresponding predictions. It should distinguish an unperformed discriminating experiment, acquired evidence omitted from an account, justified unresolved alternatives, and report-prediction inconsistency. Evaluator knowledge helps identify discriminating interventions; a claim is not wrong merely for using an equivalent representation. Independent review can assist this analysis without replacing measured predictions and retests. No completed cohort-wide mechanism score or causal compression experiment is implied.

Appendix B. Complete metric-specific arm means

Each entry is mean absolute error / empirical interval coverage (%), averaged over five world sessions. All scheduled sessions are retained, including the two source-budget shortfalls. PA uses nominal 90% intervals; all other studies use 80%. Metric scales and reference targets differ across systems. The machine-readable table includes each campaign’s interval width and protocol status.

EC, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2983 / 55.0	0.2740 / 65.0	0.1272 / 88.3
Energy efficiency	0.3098 / 33.3	0.2595 / 51.7	0.1582 / 63.3
Faradaic efficiency	0.2580 / 46.7	0.2259 / 60.0	0.1356 / 73.3
Public score	0.1953 / 51.7	0.1939 / 60.0	0.1321 / 71.7
Selective product yield	0.0731 / 60.0	0.0817 / 38.3	0.0598 / 56.7
Transport efficiency	0.2602 / 45.0	0.2184 / 66.7	0.1328 / 76.7

EC, E prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2908 / 63.3	0.2378 / 66.7	0.2661 / 73.3
Energy efficiency	0.2794 / 43.3	0.2541 / 31.7	0.2440 / 55.0
Faradaic efficiency	0.2506 / 53.3	0.2532 / 31.7	0.2315 / 56.7
Public score	0.1947 / 55.0	0.1561 / 58.3	0.1835 / 53.3
Selective product yield	0.0758 / 43.3	0.0623 / 53.3	0.0762 / 50.0
Transport efficiency	0.2478 / 58.3	0.2053 / 51.7	0.2217 / 60.0

PA, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3
Organic fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3

EC, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1902 / 68.3	0.1268 / 80.0	0.0428 / 93.3
Energy efficiency	0.1581 / 61.7	0.1651 / 68.3	0.0945 / 80.0
Faradaic efficiency	0.1458 / 65.0	0.1505 / 66.7	0.0877 / 78.3
Public score	0.1201 / 63.3	0.1452 / 73.3	0.0713 / 78.3
Selective product yield	0.0726 / 53.3	0.0573 / 70.0	0.0374 / 75.0
Transport efficiency	0.1495 / 65.0	0.1501 / 66.7	0.0846 / 78.3

EC, E prior: optimization, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1403 / 85.0	0.1252 / 85.0	0.0864 / 90.0
Energy efficiency	0.1977 / 48.3	0.1307 / 60.0	0.1779 / 65.0
Faradaic efficiency	0.1502 / 76.7	0.1098 / 78.3	0.1515 / 71.7
Public score	0.1113 / 75.0	0.0996 / 71.7	0.1136 / 75.0
Selective product yield	0.0515 / 75.0	0.0498 / 56.7	0.0587 / 58.3
Transport efficiency	0.1420 / 80.0	0.1257 / 73.3	0.1570 / 68.3

PA, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0251 / 100.0	0.0296 / 96.7	0.0207 / 100.0
Organic fraction	0.0211 / 100.0	0.0296 / 96.7	0.0207 / 100.0

RX, P prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.0800 / 88.7	0.0791 / 73.3	0.0710 / 81.0
Conversion	0.0426 / 72.7	0.0321 / 70.0	0.0306 / 73.3
Safety risk	0.1031 / 58.3	0.0282 / 85.0	0.0370 / 81.7
Public score	0.1195 / 48.7	0.0501 / 83.3	0.0476 / 90.3
Selectivity	0.1198 / 58.0	0.0818 / 69.3	0.0714 / 77.3
Yield	0.1461 / 35.7	0.0708 / 76.7	0.0698 / 78.3

RX, P prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1509 / 57.7	0.0805 / 73.0	0.1271 / 60.3
Conversion	0.0474 / 85.3	0.0319 / 82.7	0.1960 / 53.0
Safety risk	0.0984 / 56.7	0.1106 / 56.7	0.0811 / 68.3
Public score	0.1341 / 30.0	0.0818 / 73.0	0.0879 / 68.3
Selectivity	0.1900 / 31.0	0.0923 / 69.7	0.1360 / 64.7
Yield	0.2128 / 19.3	0.1060 / 65.7	0.1499 / 68.7

RX, S prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1219 / 78.7	0.1212 / 75.3	0.1279 / 64.7
Conversion	0.0217 / 97.7	0.0266 / 95.3	0.0200 / 98.0
Safety risk	0.0664 / 80.0	0.0717 / 66.7	0.1276 / 43.3
Public score	0.0504 / 89.7	0.0816 / 61.3	0.0665 / 78.0
Selectivity	0.1280 / 72.7	0.1319 / 70.7	0.1400 / 55.0
Yield	0.1157 / 81.7	0.1196 / 75.7	0.1275 / 61.0

RX, S prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1376 / 63.7	0.1490 / 67.7	0.1188 / 66.3
Conversion	0.0370 / 88.7	0.0505 / 82.7	0.0425 / 88.3
Safety risk	0.1000 / 45.0	0.1255 / 68.3	0.1133 / 45.0
Public score	0.0932 / 56.0	0.0917 / 58.0	0.0637 / 81.0
Selectivity	0.1574 / 56.7	0.1570 / 60.3	0.1175 / 62.3
Yield	0.1485 / 63.3	0.1422 / 66.3	0.0975 / 73.0

EQ, P prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0129 / 96.3	0.0759 / 68.7	0.0646 / 74.7
Normalized pH	0.0031 / 94.3	0.0160 / 65.0	0.0150 / 68.7
Precipitation signal	0.0205 / 83.3	0.0376 / 73.3	0.0383 / 68.7

EQ, S prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0292 / 91.7	0.0235 / 76.3	0.0122 / 87.7
Normalized pH	0.0097 / 87.3	0.0200 / 75.7	0.0203 / 77.7
Precipitation signal	0.0033 / 95.3	0.0042 / 95.7	0.0078 / 87.3

C, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.3279 / 25.0	0.3096 / 38.3	0.2925 / 43.3
Crystal purity	0.0095 / 98.3	0.0497 / 60.0	0.0917 / 23.3
Particle-size index	0.0777 / 55.0	0.0624 / 58.3	0.0725 / 58.3
Net crystal recovery	0.1632 / 48.3	0.0606 / 83.3	0.0892 / 71.7

C, E prior: delivery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.2281 / 51.7	0.2774 / 38.3	0.3046 / 41.7
Crystal purity	0.0197 / 95.0	0.0398 / 73.3	0.0508 / 55.0
Particle-size index	0.0566 / 71.7	0.0440 / 70.0	0.1023 / 48.3
Net crystal recovery	0.0911 / 61.7	0.0684 / 80.0	0.0786 / 73.3

P, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Purity	0.2240 / 51.7	0.1883 / 53.3	0.2508 / 50.0
Recovery	0.0668 / 56.7	0.0687 / 51.7	0.0727 / 53.3

References

- Beeler, Chris, Sriram Ganapathi Subramanian, Kyle Sprague, et al. 2024. "ChemGymRL: A Customizable Interactive Framework for Reinforcement Learning for Digital Chemistry." *Digital Discovery* 3: 742–58. <https://doi.org/10.1039/D3DD00183K>.
- Boiko, Daniil A., Robert MacKnight, Ben Kline, and Gabe Gomes. 2023. "Autonomous Chemical Research with Large Language Models." *Nature* 624: 570–78. <https://doi.org/10.1038/s41586-023-06792-0>.
- Bran, Andres M., Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. 2024. "Augmenting Large Language Models with Chemistry Tools." *Nature Machine Intelligence* 6: 525–35. <https://doi.org/10.1038/s42256-024-00832-8>.
- Elmachtoub, Adam N., and Paul Grigas. 2022. "Smart Predict, then Optimize." *Management Science* 68 (1): 9–26. <https://doi.org/10.1287/mnsc.2020.3922>.
- Felton, Kobi C., Jan G. Rittig, and Alexei A. Lapkin. 2021. "Summit: Benchmarking Machine Learning Methods for Reaction Optimisation." *Chemistry–Methods* 1: 116–22. <https://doi.org/10.1002/cmt.202000051>.
- Gandhi, Kanishk, Michael Y. Li, Lyle Goodyear, et al. 2025. *BoxingGym: Benchmarking Progress in Automated Experimental Design and Model Discovery*. <https://arxiv.org/abs/2501.01540>.
- Häse, Florian, Matteo Aldeghi, Riley J. Hickman, et al. 2021. "Olympus: A Benchmarking Framework for Noisy Optimization and Experiment Planning." *Machine Learning: Science and Technology* 2: 035021. <https://doi.org/10.1088/2632-2153/abedc8>.
- Jansen, Peter, Marc-Alexandre Côté, Tushar Khot, et al. 2024. "DiscoveryWorld: A Virtual Environment for Developing and Evaluating Automated Scientific Discovery Agents." *Advances in Neural Information Processing Systems* 37. <https://doi.org/10.52202/079017-0324>.
- Szymanski, Nathan J. et al. 2023. "An Autonomous Laboratory for the Accelerated Synthesis of Inorganic Materials." *Nature* 624: 86–91. <https://doi.org/10.1038/s41586-023-06734-w>.